Summary of Gen AI Assignment

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Applied AI / ML Engineering — Take-Home Assignment

Problem 1 — Cost-Efficient RAG Application

Background. On a fully managed vector DB, the bill scales with stored vectors (always-on pods), so a large but lightly-queried index

becomes a top infra cost. Build a working RAG app on a low-cost store and prove, with numbers, that it is a credible alternative.

Objective. A QA service over a document corpus, backed by a low-cost vector store, with honest evaluation of retrieval quality,

answer quality, latency, and cost.

Pick one store. pgvector · Qdrant (self-hosted/embedded) · ChromaDB · LanceDB · FAISS · sqlite-vec. Justify your choice;

benchmarking a second is a bonus.

Requirements

1. Ingestion of PDF/HTML/MD; configurable chunk size + overlap (state defaults); idem